

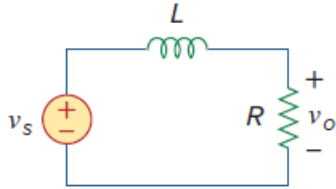
Problem

Refer to the RL circuit in Fig. 16.101. Find:

(a) the impulse response $h(t)$ of the circuit.

(b) the unit step response of the circuit.

Figure 16.101:



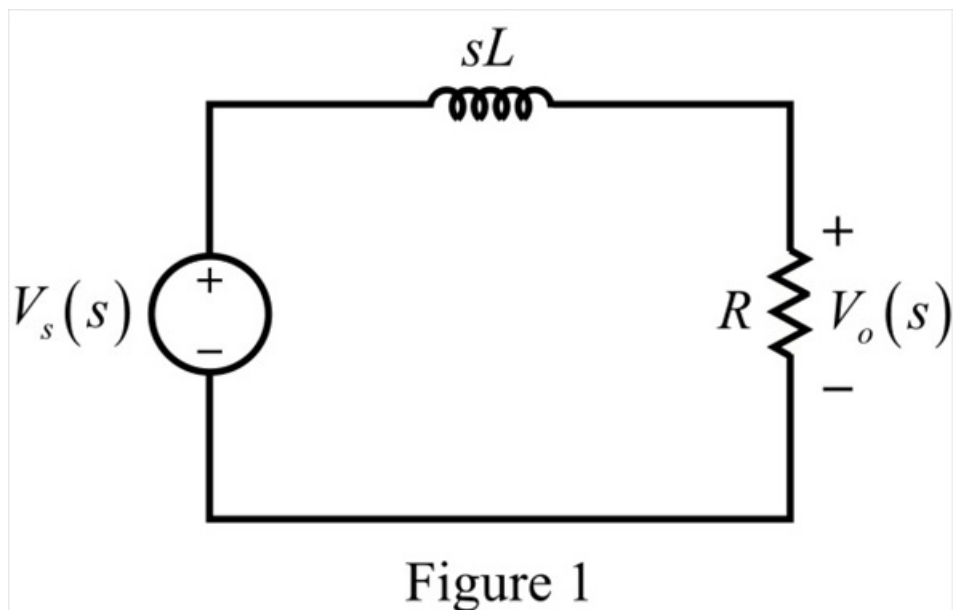
Step-by-step solution

Step 1 of 4

Refer to figure 16.101 given in the textbook.

(a)

Represent the circuit in s -domain as shown in figure 1.



Step 2 of 4

Determine the expression for the voltage across the resistor $V_o(s)$.

Apply voltage division principle.

$$V_o(s) = \frac{R}{sL + R} V_s(s) \dots\dots (1)$$

Consider the expression for transfer function $H(s)$.

$$H(s) = \frac{V_o(s)}{V_s(s)} \dots\dots (2)$$

Step 3 of 4

Consider equation (1).

Substitute the expression for $V_o(s)$ in equation (2).

$$\begin{aligned} H(s) &= \frac{\left(\frac{R}{sL+R} \times V_s(s)\right)}{V_s(s)} \\ &= \frac{R}{sL+R} \\ &= \frac{\left(\frac{R}{L}\right)}{s + \left(\frac{R}{L}\right)} \end{aligned}$$

Apply inverse Laplace transform.

$$h(t) = \frac{R}{L} e^{-\left(\frac{R}{L}\right)t}$$

Therefore, the impulse response $h(t)$ of the circuit is $\boxed{\frac{R}{L} e^{-\left(\frac{R}{L}\right)t}}$.

Step 4 of 4

(b)

Recall equation (1).

$$V_o(s) = \frac{R}{sL+R} V_s(s)$$

Substitute $\frac{1}{s}$ for $V_s(s)$.

Since, the input should be unit step for unit step response.

$$\begin{aligned} V_o(s) &= \frac{R}{sL+R} \left(\frac{1}{s}\right) \\ &= \frac{\frac{R}{L}}{s \left(s + \frac{R}{L}\right)} \\ &= \frac{\left(s + \frac{R}{L}\right) - s}{s \left(s + \frac{R}{L}\right)} \\ &= \frac{1}{s} - \frac{1}{\left(s + \frac{R}{L}\right)} \end{aligned}$$

Apply the inverse Laplace transform.

$$\begin{aligned} v_o(t) &= u(t) - e^{-\left(\frac{R}{L}\right)t} u(t) \\ &= \left[1 - e^{-\left(\frac{R}{L}\right)t}\right] u(t) \text{ volts} \end{aligned}$$

Therefore, the unit step response of the circuit is $\boxed{\left[1 - e^{-\left(\frac{R}{L}\right)t}\right] u(t) \text{ volts}}$.